

Chapter 3: Nutrition During Pregnancy

At the end of this session you will be able to:

- Describe the patterns of fetal growth and development.
- Discuss the normal physiological changes occur in pregnant women.
- Identify the nutrient requirements by and recommendations for pregnant women.
- Identify the common health problems during pregnancy.

Introduction

- ✓ Pregnancy occurs when a male's sperm reaches in the uterus of a woman, meets and fertilizes the woman's ovum
- ✓ Gestation period is The period from fertilization to birth
- ✓ Pregnancy lasts from 38 to 42 weeks, (40 weeks on the average).
- ✓ Pregnancy is a period of rapid growth and development of the fetus, and high physiological and metabolic demand on the mother.
- ✓ Maternal diet and nutritional status have a direct impact on the course of pregnancy and its outcome.
- ✓ Malnutrition that occurs in the early months of pregnancy affects development and the capacity of the embryo to survive whilst poor nutrition in the latter part of pregnancy affects fetal growth

Fetal Growth and Development

- ✓ The rate of human growth and development is higher during gestation than at any time.
- ✓ Fetal Body Composition
 - ✓ The fetus undergoes marked changes in body composition during pregnancy i.e. a progressive increases in fat, protein, and mineral content.
 - ✓ Some of the most drastic changes take place in the last 5 weeks of pregnancy, when fat and mineral content increase substantially.

Fetal Growth and Development

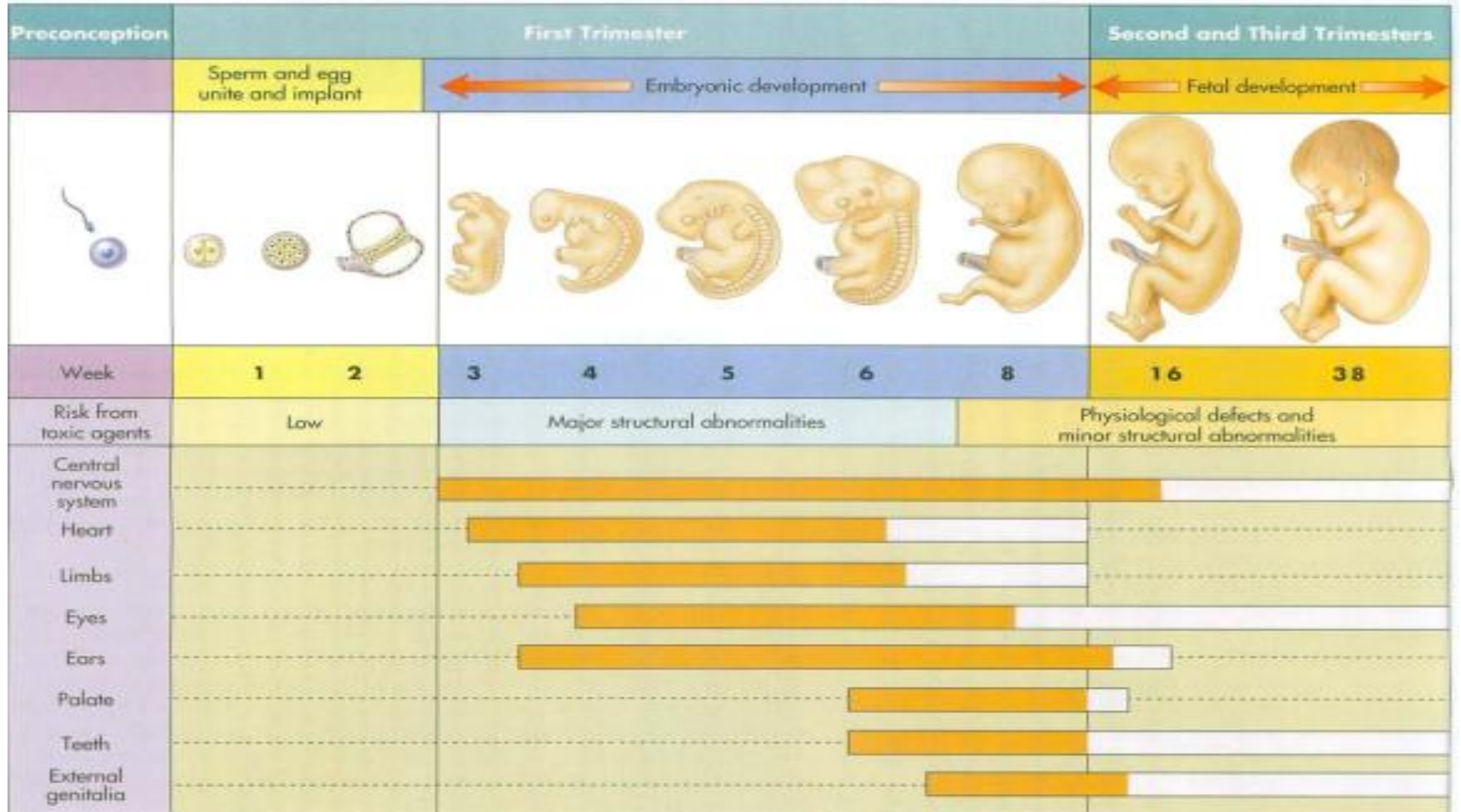
- Fetal growth is divided into three stages:
 - **Blastogenesis stage (two-weeks)**: where the fertilized ovum rapidly divides and implants itself in the endometrial lining of the uterus.
 - **Critical embryonic stage**: where the rudiments of all the principal organs and membranes develop and;
 - **Fetal stage**: which extends from the third month until term.
- Immediately after fertilization takes place, cell division begins and progresses at a rapid rate.

Fetal Growth and Development

- ❖ During the embryonic stage the fetus is particularly vulnerable to retarded development or abnormality if
 - necessary nutrients, like; folic acid, are not available or
 - exposed to teratogens such as drugs, alcohol, or excessive vitamin A.
- ❖ Any abnormal development of the neural tube or its failure to close completely can cause a major defect in the central nervous system.

- During embryonic development (from 2 to 8 weeks), many of the tissues are in their critical periods (yellow area of the bars); events occur that will have irreversible effects on the development of those tissues.
- In the later stages of development (white area of the bars), the tissues continue to grow and change, but the events are less critical in that they are relatively minor or reversible.

Fetal Growth and Development



Normal Physiological Changes During Pregnancy

- Physiological changes in pregnancy can be divided into two basic groups:
 - Those occurring in the first half of pregnancy
 - ✓ (10% of fetal growth is accomplished):
 - ✓ “maternal anabolic” changes because they build the capacity of the mother’s body to deliver relatively large quantities of blood, oxygen, and nutrients to the fetus in the second half of pregnancy, and;
 - Those occurring in the second half of pregnancy
 - ✓ (90% of fetal growth occurs)
 - ✓ “maternal catabolic” changes in which energy and nutrient stores, and the capacity to deliver stored energy and nutrients to the fetus, predominate.

Table 4.7 Summary of maternal anabolic and catabolic phases of pregnancy¹¹⁻¹³

Maternal Anabolic Phase 0–20 Weeks	Maternal Catabolic Phase 20+ Weeks
Blood volume expansion, increased cardiac output	Mobilization of fat and nutrient stores
Buildup of fat, nutrient, and liver glycogen stores	Increased production and blood levels of glucose, triglycerides, and fatty acids; decreased liver glycogen stores
Growth of some maternal organs	Accelerated fasting metabolism
Increased appetite, food intake (positive caloric balance)	Increased appetite and food intake decline somewhat near term
Decreased exercise tolerance	Increased levels of catabolic hormones
Increased levels of anabolic hormones	

Normal Physiological Changes...(1)

► Changes in the gastrointestinal tract

- ✓ Increased **progesterone production** leads to decreased **tone and motility of the smooth muscle** of the gastrointestinal tract.
- ✓ Food moves through the GI tract more slowly which leads to **constipation** and women also experience **nausea and vomiting**

► Increase in blood volume

- ✓ Blood volume increases by 50% so that enough oxygen can be available to support growing maternal and fetal tissues.
- ✓ Increase total body water throughout pregnancy.

► Increased need for those nutrients most important for generating blood:

- ✓ protein, iron folic acid and vitamin B6

► Changes in weight

- ✓ The woman gains weight as both her tissues and the fetal tissues increase and develop

Normal Physiological Changes...(2)

➤ **Altered renal function**

- ✓ Glomerular filtration rate is increased to facilitate clearance of urea and other waste products from the mother and the fetus.

➤ **Increase in breast tissue and fat stores**

- ✓ The mammary glands in the breast begin to enlarge during pregnancy to allow the milk ducts to prepare for lactation.
- ✓ Fat stores increase in preparation for breast feeding

➤ **Increase the basal metabolic rate**

Normal Physiological Changes...(3)

► **Metabolic changes and adaptive responses**

❖ Changes in maternal hormone secretions lead to

- ✓ Changes in the utilization of carbohydrate, fat and protein during pregnancy.
The fetus requires a continual supply of glucose and amino acids for growth.
- ✓ Maternal tissues to make greater use of lipids for energy production.
- ✓ Increased absorption of iron and reduced urinary excretion of riboflavin
- ✓ Increased secretion of aldosterone, at end of pregnancy, --- increased reabsorption of sodium from the renal tubules

placenta

- Primary function is to act as organ of respiration and to be the pathway for **exchange of nutrients** and **waste products between** mother and fetus.
- To accomplish these tasks , it uses all absorption mechanisms that employed by GI tract.
- **Functions of the placenta include:**
 - ✓ Hormone and enzyme production,
 - ✓ Nutrient and gas exchange between the mother & fetus
 - ✓ Removal of waste products from the fetus
 - ✓ A barrier to the passage of maternal red blood cells, bacteria, and many large proteins.
 - ✓ prevents the mixing of fetal and maternal blood until delivery, when ruptures in blood vessels may occur.

Pregnancy Weight Gain

- Weight gain during pregnancy is an important consideration because newborn weight and health status tend to increase as weight gain increases.
 - Too small weight gain is associated with reduced fetal growth and LBW or small for gestational age infants.
 - Excessive gain--infants (>4200gm), increased likelihood of caesarean delivery, and post-partum obesity.

- ✚ Pre pregnancy weight status influences the relationship between weight gain and birth weight.
- ✚ A normal weight gain for most healthy women is between 11 and 15 kg, and on average about 12.5 kg.

Table 4.17 Pregnancy weight gain recommendations⁷⁵

Prepregnancy Weight Status Body Mass Index	Recommended Weight Gain
Underweight, <18.5 kg/m ²	28–40 lb (12.7–18.2 kg)
Normal weight, 18.5–24.9 kg/m ²	25–35 lb (11.4–15.9 kg)
Overweight, 25–29.9 kg/m ²	15–25 lb (6.8–11.4 kg)
Obese, 30 kg/m ² or higher	11–20 lb (5.0–9.1 kg)
Twin pregnancy	25–54 lb (11.4–24.5 kg)

Components of weight gain during pregnancy



First



Second



Third

**Total weight gain
25–35 lbs**

- Maternal fat stores (4–11 lbs)
- Uterus and breast (4 lbs)
- Blood (3–4 lbs)
- Fetus (7–8 lbs)
- Placenta, amniotic fluid, and other fluids (7–8 lbs)

Nutrient requirements during pregnancy

- Nutrient requirements are vary during the course of pregnancy depending on:
 - ✓ Pre-pregnancy nutrient stores,
 - ✓ Physical activity levels,
 - ✓ Stage of pregnancy and health status.
- For the most part, nutrient needs can be optimally met by consuming well balanced, adequate and healthful diets consisting of basic foods.

Calories

- ❖ Energy requirements increase during pregnancy, mainly due to increased maternal body mass and fetal growth.

- ❖ Energy requirement during pregnancy:

 - ✓ 2nd trimester: + 340 kcal/day

 - ✓ 3rd trimester: + 450 kcal/day .

- ❖ Calorie needs increase because of:

 - ✓ The increase in basal metabolic rate.

 - ✓ Weight gain increases the amount of calories burned during activity.

 - ✓ Uses additional calories to store energy in preparation for lactation after delivery

2. Carbohydrate Needs During Pregnancy

- Approximately 50–60% of total caloric intake
- Consume a minimum of 175 g CHO to meet the fetal brain's need for glucose
- Basic foods such as vegetables, fruits, and whole-grain products containing fiber and a variety of other nutrients
- - ✓ They also protect pregnant women from constipation.

3. Protein Needs During Pregnancy

- The recommended is **+25 g/ day**, or **71** grams daily, and **1.1** gram/ kg body weight.
- Protein requirements increase during pregnancy primarily due to **protein tissue accretion and development**.
- approximately 925 grams of protein accumulated in protein tissues during pregnancy:
- Protein needs increase to support :
 - Fetal growth and development.
 - The formation of the placenta and amniotic fluid.
 - The growth of maternal tissues and the expanded blood volume.

4. Fat Needs During Pregnancy

- Consume 33 % of total calories on average from fat.
- Fat consumed in foods is used as:
 - ✓ energy source & provides essential fatty acids required for fetal growth and development
 - ✓ a source of fat-soluble vitamins.
- It is recommended that pregnant women consume
 - ✓ 13 grams of Linoleic acid daily, and
 - ✓ 1.4 grams of alpha Linolenic acid.

Vitamins and Minerals Need During Pregnancy

- Requirements for most vitamins and minerals increase during pregnancy due to metabolic demands associated with placental and fetal growth, expansion of maternal tissues and plasma volume, and increased nutrient needs for tissue maintenance.
- Maternal physiological adaptations involve changes in vitamin and mineral absorption and utilization that respond to the changing needs for these nutrients by time in pregnancy.

Vitamin Needs During Pregnancy

Folate

- Folate requirements increase dramatically during pregnancy due to the extensive organ and tissue growth that takes place.
- Inadequate folate during pregnancy has long been associated with:
 - ✓ anemia
 - ✓ reduced fetal growth.
 - ✓ development of congenital abnormalities.
- enzyme cofactor in metabolic reactions involved in the synthesis of DNA, gene expression & regulation.
- Deficiency of folate impairs these processes, leading to abnormal cell division and tissue formation.

Folic acid

- ✓ 600 mcg daily once pregnancy is confirmed.
- ✓ Women who consume adequate amounts of folic acid before conception and throughout the first month of pregnancy reduce their risk of having a baby with a neural tube defect (e.g., spina bifida, anencephaly).



Other B vitamins

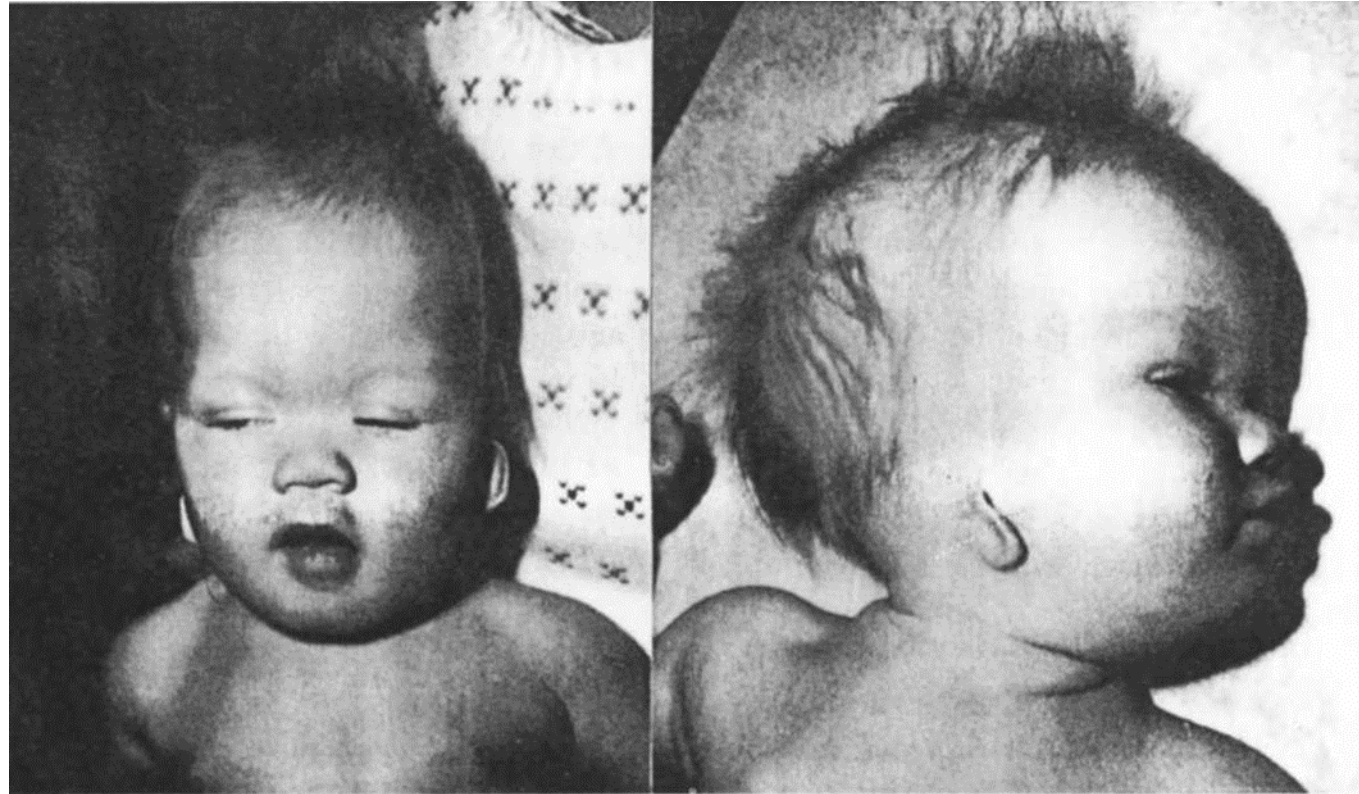
- ❖ The increased requirement for vitamin B6 is proportional to the increase in protein because it is involved in protein metabolism.
- ❖ Because vitamin B12 is necessary for the metabolism of folate, a slight increase in intake is recommended.

Vitamin A

- ✓ Vitamin A is a key nutrient in pregnancy because it plays important roles in reactions involved in cell differentiation.
- ✓ RDA 1300mcg/d
- Vitamin A deficiency that occurs early in pregnancy can produce malformations of the fetal lungs, urinary tract, and heart.
- Due to the potential toxicity women take no more than 5,000 IU of vitamin A as retinol from supplements during pregnancy.
 - High intakes of beta-carotene have not been related to birth defects.

An infant exposed to high levels of retinoic acid *in utero*.

Note: high forehead, flat nasal bridge, and malformed ear.



Vitamin D

- ✓ Vitamin D supports fetal growth, the addition of calcium to bone, and tooth formation.
 - ✓ Lack of vitamin D during pregnancy compromises fetal as well as childhood bone development.
- ✓ Infants born to women with poor vitamin D status tend to be
 - ✓ smaller than average,
 - ✓ more likely to have low blood calcium levels (hypocalcaemia) at birth
 - ✓ more likely to have poorly calcified bones and abnormal enamel & develop dental caries in childhood.
- ✓ Vitamin D also supports normal functioning of the immune system and can inhibit inflammation.
- ✓ An intake of 5 mcg (200 IU) vitamin D daily along with exposing the skin to sunshine.

Need for Minerals During Pregnancy

Calcium

- needed for fetal skeletal mineralization and maintenance of maternal bone health.
- Fetal demand for calcium peaks in the **third trimester** when fetal bones are mineralizing at a high rate.
 - The AI for is 1000 mg

Iron

❖ Recommended Intake of Iron for women

- ✓ DRI of iron for non-pregnant women (18 mg/day).
- ✓ RDA of iron for pregnancy is 27 mg per day.
- ✓ The UL of iron during pregnancy is 45 mg per day.

❖ Women require about 1000 mg (1 g) of additional iron for pregnancy.

- ✓ 300 mg for fetus & placenta
- ✓ 250 mg lost at delivery
- ✓ 450 mg for increased RBC

Importance for iron during pregnancy:

- ✓ To support the increase in maternal blood volume.
- ✓ To provide iron for fetal liver storage, which will sustain the infant for the first 4 - 6 months of life.

Iodine

RDA=220mcg for pregnant, 150mcg for non pregnant, UL=1100mcg.

- Iodine is required by the mother and fetus for **thyroid function, energy production**, and for **fetal brain development**.
- Deficiency of iodine early in pregnancy can lead to **hypothyroidism** in the offspring.
 - Supplements should be given before or in the first half of pregnancy.
- Women who consume iodized salt are not likely to need supplemental iodine.
 - The most reliable source of iodine is iodized salt 1(teaspoon contains 400 mcg iodine).
 - Fish, shellfish, seaweed, and some types of tea provide iodine

DRIs for Pregnant and Nonpregnant Women Aged 19-30 years

Energy, kcal			
2nd trimester	+350	2403	—
3rd trimester	+452		
Protein, gm	71	46	—
Linoleic acid, g	13	12	—
Alpha-linolenic acid, g	1.4	1.1	—
Vitamin A, mcg	770	700	3000
Vitamin C, mg	85	75	2000
Vitamin D, mcg ^a	5	5	50
Vitamin E, mg	15	15	1000 ^c
Vitamin K, mcg	90	90	—
Thiamin, mg	1.4	1.1	—
Riboflavin, mg	1.4	1.1	—
Niacin, mg	18	14	35 ^c
Vitamin B ₆ , mg	1.9	1.3	100
Folate, mcg ^b	600	400	1000 ^{c,d}
Vitamin B ₁₂ , mcg	2.6	2.4	—
Pantothenic acid, mcg	6	5	—
Biotin, mcg	30	30	—
Choline, g	450	425	3.5
Calcium, mg	1000	1000	2500
Chromium, mcg	30	25	—
Copper, mcg	1000	900	10,000
Fluoride, mg	3	3	10
Iodine, mcg	220	150	1100
Iron, mg	27	18	45
Magnesium, mg	350	310	350 ^c
Manganese, mg	2	1.8	11
Molybdenum, mcg	50	45	2000
Phosphorus, mg	700	700	3500
Selenium, mcg	60	55	400
Zinc, mg	11	8	40

Sodium

- Sodium plays a critical role in maintaining the body's water balance.
- Inadequate sodium intake can complicate the course and outcome of pregnancy.
- Requirements for it increase markedly during pregnancy due to plasma volume expansion.

Common Nutrition-Related Concerns of Pregnancy

Nausea and Vomiting

- Called “**morning sickness**” since they occur mostly after waking up.
- Symptoms of nausea generally begin around week 5 of gestation and disappear by week 12.
 - They are so common *i.e.* considered as a normal part of pregnancy.
- The cause of nausea and vomiting is not yet clear, but they are thought to be due to the increased levels of human chorionic gonadotropin, progesterone, estrogen, or other hormones early in pregnancy.
 - Iron supplements may aggravate nausea and vomiting when taken in the first trimester of pregnancy.

Management

- ❖ Eating small, frequent meals every 2 to 3 hours.
- ❖ Eat carbohydrate foods such as: dry crackers, dry cereal,.
- ❖ Take Vitamin B6 (pyridoxine) supplements
- ❖ Ginger 1gram a day for 4 days may decrease nausea and vomiting
- ❖ Avoid odors and foods that trigger nausea.
- ❖ Limit high-fat foods
- ❖ Avoid coffee, tea and spicy foods, Avoid drinking liquids with meals.

2. Heartburn: . The hormones of pregnancy relax the digestive muscles, and the growing fetus puts increasing pressure on the mother's stomach.

This combination allows stomach acid to back up into the lower esophagus, creating a burning sensation near the heart

Management :

- ✓ Eat small, frequent meals.
- ✓ Avoid coffee, high-fat foods and spices.
- ✓ Avoid lying down or bending over after eating.

3. Constipation

May be caused by:

- ✓ Relaxation of gastrointestinal muscle tone and motility
- ✓ Pressure of the fetus on the intestines.
- ✓ Decrease in physical activity
- ✓ Inadequate intake of fluid and fiber.
- ✓ Side effect of the consumption of iron supplements.

Management:

- ✓ Increase fiber intake.
- ✓ Drink at least eight 8- glasses of liquid daily.
- ✓ Try hot water with lemon or prune juice upon waking to help stimulate peristalsis
- ✓ Participate in regular exercise.

Food Cravings and Aversions Some women develop cravings for, or aversions to, particular foods and beverages during pregnancy.

food cravings: strong desires to eat particular foods.

food aversions: strong desires to avoid particular foods.

- ✓ Most likely they occur due to hormone-induced changes in sensitivity to taste and smell.

Nonfood Cravings Some pregnant women develop cravings for nonfood items such as freezer frost, laundry starch, clay, soil, or ice—a practice known as pica.

Food Cravings and Aversions

Food Cravings and Aversions Some women develop cravings for, or aversions to, particular foods and beverages during pregnancy.

food cravings: strong desires to eat particular foods.

food aversions: strong desires to avoid particular foods.

✓ Most likely they occur due to hormone-induced changes in sensitivity to taste and smell.

Nonfood Cravings Some pregnant women develop cravings for nonfood items such as freezer frost, laundry starch, clay, soil, or ice—a practice known as pica.

➤ The specific craving for nonfood items that come from the earth, such as clay or dirt, is known as *geophagia*.

Nutritional interventions for medical complications during pregnancy

1. Diabetes mellitus

- It characterized by abnormal glucose tolerance, requires dietary management regardless of whether it was present before conception (established diabetes) or developed during gestation (gestational diabetes) as a result of the metabolic changes of pregnancy.

Diabetes increases the risk of:

- ✓ urinary tract infection.
- ✓ Preeclampsia & Eclampsia
- ✓ Spontaneous abortion
- ✓ Extrauterine conception
- ✓ Neonatal death
- ✓ Congenital abnormalities

- **Gestational diabetes:** high blood glucose levels develop after about twentieth week of pregnancy
- Can result in **macrosomia:** large baby weighing more than 8 lbs
- Increases risk of baby having **jaundice** (yellowish coloring of the skin due to the presence of bile pigments in the blood), breathing problems, birth, defects

❖ Gestational diabetes does not usually produce maternal complications or birth defects, but it can make delivery difficult, because babies born to gestational diabetics are usually large , which may increase the risk of postpartum hemorrhage.

Diabetes Management:

- Monitor the progress and course of pregnancy of established diabetics.
- Screen all women for gestational diabetes between 24 - 28 weeks of pregnancy.
- Check for ketonuria regularly.
- Diabetic management during pregnancy includes nutrition therapy and, possible, multiple daily doses of insulin.

Advise the client that:

✓ Pregnant diabetics require the same nutrients and weight gain as non diabetic pregnant women.

dietary interventions

❖ high pre-pregnancy (BMI) and excessive weight gain during pregnancy increases the risk of GDM, pre-eclampsia.

- lifestyle change interventions
 - (diet and exercise) reduce the risk of excessive GWG
 - calorie restriction for overweight and obese women with GDM to avoid excessive GWG

Hypertensive disorder of pregnancy

- ✓ Pregnancy–induced hypertension (PIH or toxemia) is a hypertensive syndrome that occurs in approximately 6 - 7% of all pregnancies.
- ✓ Elevated blood pressure (systolic ≥ 140 or diastolic ≥ 90 mm Hg,
- ✓ It includes : chronic hypertension gestational hypertension, preeclampsia, eclampsia
- ✓ Severe cases are associated with increased risks of maternal, fetal and neonatal death

- ✓ **Gestational hypertension:** high blood pressure develops about halfway through pregnancy (stress on kidneys and heart)
- ✓ **Preeclampsia:** includes hypertension, severe edema and protein in urine
 - Treatment includes bed rest, medication, even hospitalization until baby can be safely delivered
 - If untreated, can lead to eclampsia
- ✓ **Eclampsia:** can cause seizures in mother and is major cause of death of women during pregnancy

Management:

- Lifestyle interventions,
-a reduction in salt intake

Maternal factors that affect pregnancy outcome

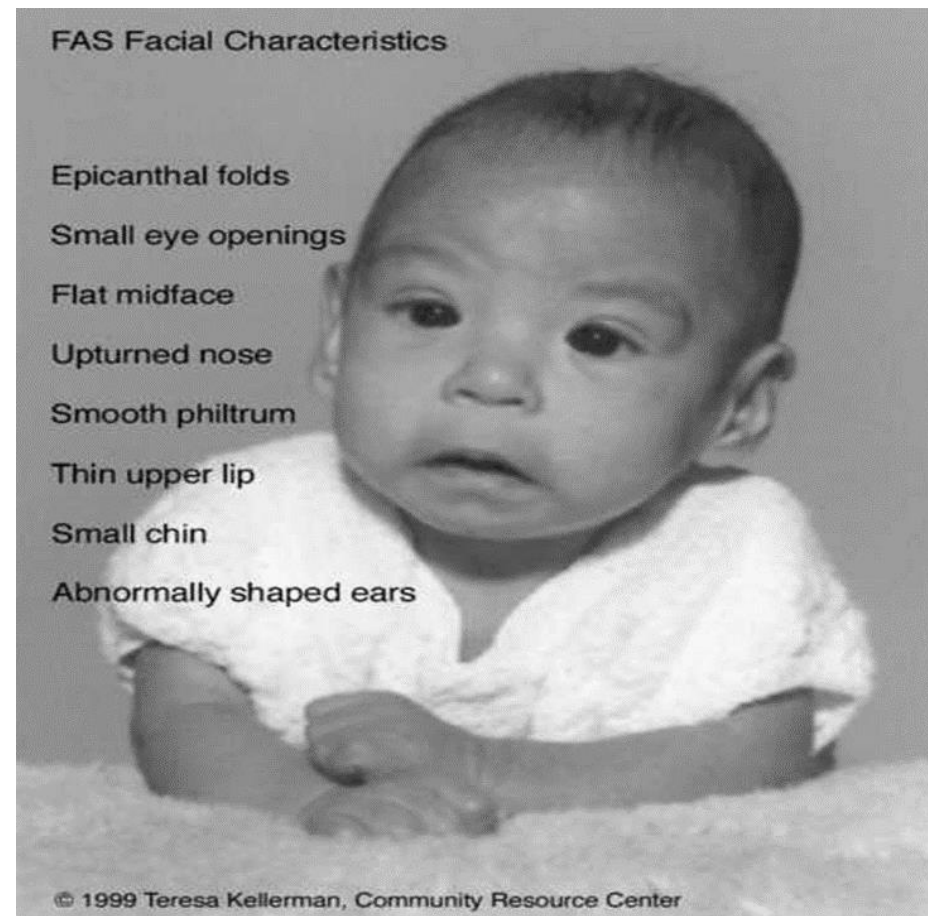
- **Pre-pregnant Weight**
- **Maternal Age**
- **Weight gain during pregnancy**

Lifestyle factors that impact on pregnancy Outcome

Alcohol Consumption and Drug use

- The most serious fetal damage can occur in the early embryonic stage.
- Heavy alcohol passes through the placenta to the fetus & increases the **risk of miscarriage, stillbirth, and infant death** (within the first month after delivery).
- Alcohol exposure during critical periods of growth and development can permanently impair organ and tissue formation, growth, health, and mental development in the offspring.

Frequent consumption of high amounts of alcohol from early pregnancy onward is related to the development of **fetal alcohol syndrome**.



Lifestyle factors that impact on pregnancy Outcome

Coffee

- ✓ The foetus has a limited ability to metabolize caffeine which is known to cross the placenta.

3. Smoking

- ✓ causes retardation of fetal growth & risk of LBW baby.

4. Exercise

- ✓ moderate exercise benefits the mother but hard physical exercise affect fetal development & results in LBW babies.

Chapter 4

Nutrition During Lactation



Objectives

At the end of the this session you will be able to:

- Explain the physiology of lactation.
- Discuss the human milk composition.
- Identify composition of breast milk
- Describe importance of breast feeding
- Discuss the energy and nutrient requirements during Lactation.

Introduction

- ❖ The health care system, the workplace, and the community can either hinder or facilitate the initiation and continuation of breastfeeding.
- ❖ Health care professionals who wish to manage and promote breastfeeding should understand the physiology of lactation, the composition of human milk, and the benefits to mothers and infants.

Key Nutrition Concet

1. Human milk is the best food for newborn infants for the first year of life or longer.
2. Maternal diet does not significantly alter the protein, carbohydrate, fat and major mineral composition of breast milk, but it does affect the fatty acid profile and the amounts of some vitamins and trace minerals.
3. When maternal diet is **inadequate**, the quality of the milk is preserved over the quantity for the majority of nutrients.

Physiology of Lactation

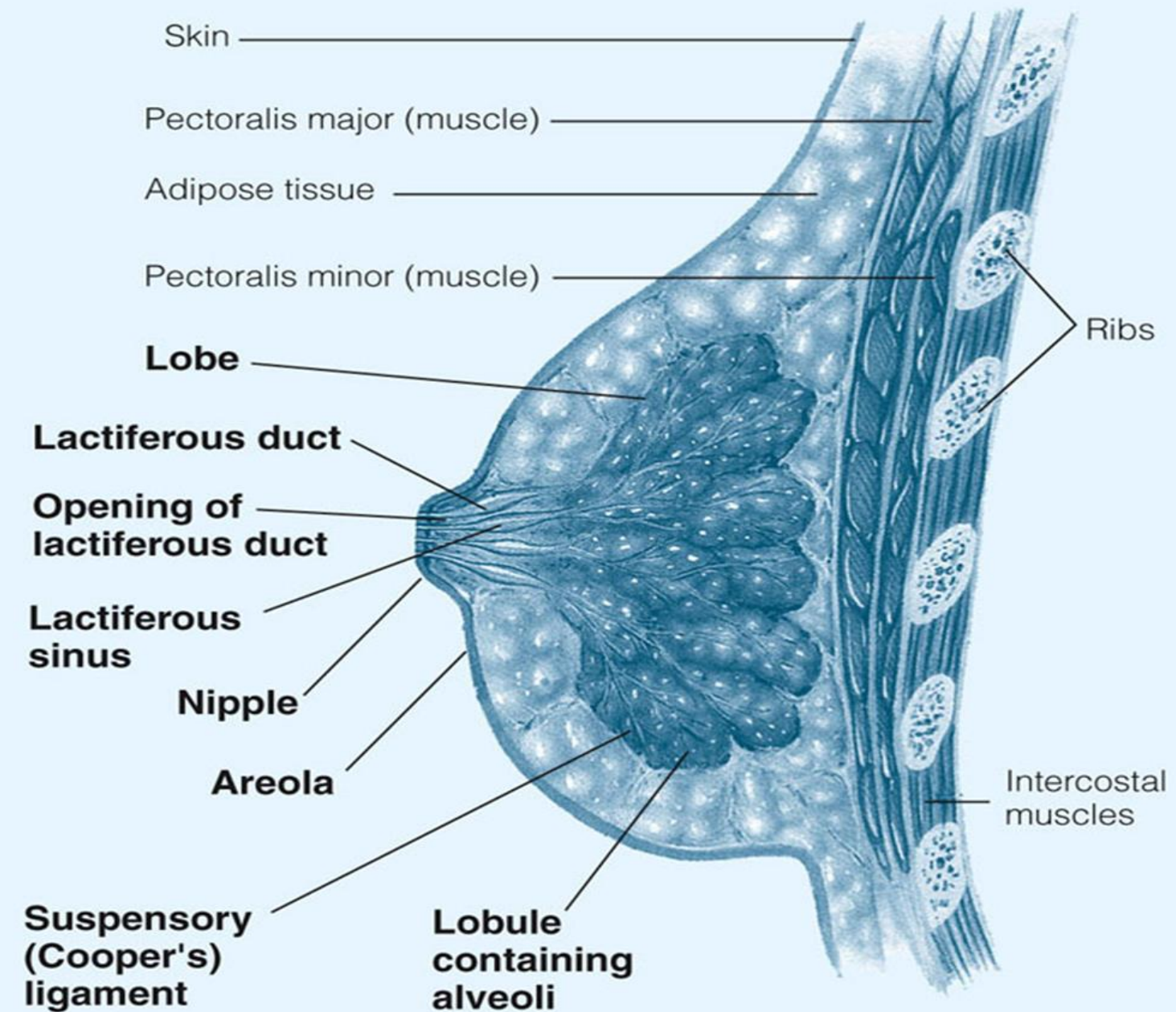
Lactogenesis

- it is regulation of breast milk secretion (production); it occurs in three stages.
- 1. **Lactogenesis I**: begins during the last trimester of pregnancy and extends through the first few days postpartum.
 - ✓ Milk begins to form, the lactose and protein content of milk increase.
- 2. **Lactogenesis II** (“milk comes in”): begins 2–5 days postpartum and is marked by increased blood flow to the mammary gland.
- 3. **Lactogenesis III**: begins about 10 days after birth and is the stage in which the milk composition becomes stable.

Hormonal Control of Lactation

- **Prolactin**
 - Stimulates milk production
 - Released in response to suckling,
 - levels double with suckling
- **Oxytocin**
 - Stimulates letdown/ ejection of milk from milk gland (acinus) into the milk ducts.
 - Tingling of the breast may occur corresponding to contractions in milk duct
 - Causes uterus to contract, seal blood vessels, & shrink in size

Breast of a Lactating Female



Human Milk Composition

- Human milk is the only food needed by the majority of infants for approximately 6 months.
- Human milk have hundreds of identified components.
- The composition of milk is designed not only to nurture, but also to protect infants from infectious and certain chronic diseases.
- Depending upon the stage of lactation, breast milk is designated as:
 1. ***Colostrum*** (3–5 days post partum)
 2. ***Transitional milk*** (5–15 days)
 3. ***Mature milk*** (secreted after 15 days or 2 weeks of infant birth).

Human Milk Composition...

❖ Colostrum

- ✓ The first milk
- ✓ Is a thick, often yellow fluid produced (days 1–3 after infant birth).
- ✓ It provides about 580–700 kcal/l and is higher in protein and lower in carbohydrate & fat than mature milk.
- ✓ Secretory IGA and lactoferrin are the primary proteins present in it
- ✓ Has higher concentrations of (white blood cell provides immune protection), sodium, potassium, and chloride than mature milk.

Table 6.4 Compositions of 100 mL colostrum (days 1–5 postpartum) and mature milk (day 15 postpartum)

Contents	Colostrum	Mature Milk
Calories (kcal)	55	67
Fat (g)	2.9	4.2
Lactose (g)	5.3	7.0
Total protein (g)	2.0	1.1
Secretory IgA	0.5 ^a	0.1
Lactoferrin	0.5	0.2
Casein	0.5	0.4
Calcium (mg)	28	30
Sodium (mg)	48	15
Vitamin A (µg retinol equivalents)	151	75
Vitamin B ₁ (µg)	2	14
Vitamin B ₂ (µg)	30	40
Vitamin C (µg)	6	5

^aConcentration is considerably higher at 1–3 days postpartum than at days 4 and 5.

SOURCE: Adapted from Prentice A. Constituents of human milk. *Food and Nutrition Bulletin* 17(4), The United Nations University Press, December 1996.¹⁶

Human Milk Composition...

Water

- Breast milk is isotonic like plasma (has balanced concentration of salts and minerals).
 - Breast feeding babies do not need water or other fluids to maintain hydration, even in hot climates.

Energy

- ✓ ~0.65 kcal/mL
- ✓ Lower in calories than human milk substitute (HMS)

Human Milk Composition...

Protein (Casein, Whey Proteins)

- The protein content of mature human milk is relatively low (0.8 to 1%) compared to cow's milk.
- Casein
 - Main protein in mature human milk
 - Facilitates calcium absorption
- Whey
 - Some minerals, hormones & vitamin binding proteins are part of whey

Human Milk Composition...

Lipids & Trans Fatty Acids

- Lipids provide half of the energy of human milk.
- Human milk contains omega-3 fatty acids
- DHA (docosahexaenoic acid)
 - ✓ Essential for retinal development
 - ✓ Associated with **higher IQ** score

Cholesterol

- Is an essential component of all cell membranes, and needed for growth and replication of cells.
- Higher in human milk than HMS
- Early consumption of cholesterol through breast milk appears to be related to lower blood cholesterol levels later in life

Human Milk Composition...

Carbohydrates

- Lactose is the dominant carbohydrate in human milk that enhances calcium absorption.
- Monosaccharides (glucose), polysaccharides & oligosaccharides (stimulate the growth of Bifidus bacteria in the gut, inhibit the growth of *E. coli* and other potentially harmful bacteria).

Fat-Soluble Vitamins (Vitamin A, D, E & K)

Water-Soluble Vitamins (vitamin C, B2, B3,

- ✓ Content *reflective of mother's diet*
- ✓ B6 (most likely to be deficient),
- ✓ biotin, B12 and Folic Acid (bound to whey proteins)).

Human Milk Composition...

- + **Minerals** (magnesium, calcium, iron, zinc) in human milk are bioavailable to the infant:.
- + **Trace Minerals** (copper, selenium, chromium, manganese, molybdenum, nickel & fluorine) are essential for growth and development.

bioavailability

	iron	zinc
Human milk	49%	41%
Cow milk	4%	31%
Fortified milk	10%	28%

Benefits of Breastfeeding

Benefits of Breastfeeding for Mothers

- ✓ return uterus to non pregnant size, minimizes maternal postpartum blood loss & to expel placenta.
- ✓ Natural contraceptive
- ✓ Psychological benefits (facilitate bonding with their infants).
- ✓ Help the mother to lose excessive weight.
- ✓ Reduce the risk of breast and ovarian cancer.
- ✓ Economic benefits, breast milk is (almost) free, available at all times.

Benefits of Breastfeeding

Breastfeeding Benefits for Infants

❖ **Nutritional Benefits**: human milk contains balanced nutrients.

- ✓ Meets the infant's needs without overloading the immature kidneys with nitrogen.
- ✓ Whey protein in human milk forms a soft, easily digestible curd.
- ✓ Provides essential fatty acids like: long-chain PUFA (DHA), which promotes optimal development of the central nervous system.

Benefits of Breastfeeding

- **Immunological Benefits:** protect against infections.
 - Breast milk contains
 - cells (T- and B-lymphocytes), IgA, Bifidus factor, carrier proteins (lactoferrin, vitamin B12-binding protein, and corticoid-binding protein), and
 - enzymes (lysozyme, lipoprotein lipase, leukocyte enzymes) confer immunological benefits.
- **Reductions in acute & Chronic Illness** (celiac disease, inflammatory bowel disease, allergy and asthmatic disease, ear infection).
- **Reduce the risk of childhood Overweight**
- **Cognitive Benefits:** cognitive development increases with the duration of breastfeeding.

Benefits of Breastfeeding

- **Socioeconomic Benefits:**
 - Decrease in medical care .
 - Reduce the time lost due to infant morbidity and mortality.
 - Improve productivity.

Energy and Nutrient Needs for Lactation

- The DRIs for normal-weight lactating women assume that the energy spent for milk production is 500 Kcal per day in the first 6 months and 400 Kcal afterward.
- there should be a substantially greater dietary energy and nutrient intake during lactation.

⚙ **WHO/FAO recommendations:**

- If a mother delivers 850 ml of milk per day for six months, around 550 kcal/day of additional energy is required to make that milk.
 - The average energy requirements for moderately active woman is 2,200 kcal/day, the total recommendation becomes 2,750 kcal/day.

Maternal Energy Balance and Milk Composition

- Protein–calorie malnutrition results in energy deficiency and can reduce the volume of milk produced but does not compromise the composition of the milk.

Protein

- On the basis of an 850 ml output, assuming it contains 1.2gram of protein per 100 ml, an extra 24 gram/day of protein with a chemical score of 70gram/day was recommended for lactation by FAO/WHO.

Vitamins and Minerals Supplement

- Well-nourished breastfeeding women do not need routine vitamin or mineral supplementation.
- Instead, supplementation should target specific nutritional needs of individual women. :
 - ✓ In the cases of proven dietary inadequacies, **vegan women** require vitamin B12 supplementation.
 - ✓ When **exposure to sunlight** is considered inadequate, maternal vitamin D supplementation may be advisable.
 - ✓ Women who avoid **dairy products** completely should use calcium (1200 mg) and vitamin D (10 µg) supplements

- The requirements for most are similar to requirements during pregnancy; except an increased requirement for vitamin A, calcium, vitamin C, and zinc.

Weight Loss During Breastfeeding

- Women who consumed fewer than 1500 Kcal tended to experience a decrease in milk volume.
- On the other hand women who consumed over 1500 Kcal per day experienced no decrease.